Test 217:

1) _ _ _ _ $1, 2, 3, 4, 5$

$4 \cdot 4 \cdot 4 \cdot 4 = 256$

$\underline{5} \cdot \underline{5} \cdot \underline{5} \cdot \underline{5} = 625 - 256 = 369$ (R: E)

2) 7 membri Case 1: P - Amn

6 + Ama

$4 \cdot (6 \cdot 5 \cdot 4) = 480$

$7^{\underline{4}} =$

$7^{\underline{4}} = \frac{7 \cdot 6 \cdot 5 \cdot 4}{1 \cdot 2 \cdot 3 \cdot 4} = 35$ $A = \frac{n!}{(n-k)!}$

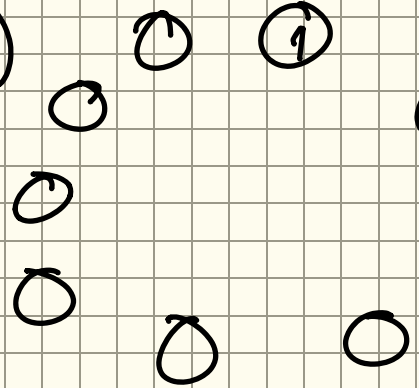
$6^{\underline{4}} = \frac{6 \cdot 5 \cdot 4 \cdot 3}{1 \cdot 2 \cdot 3 \cdot 4} = 15$ $= \frac{7!}{3!} = 7 \cdot 6 \cdot 5 \cdot 4$

$= 30 \cdot 28 = 840$

$840 - 360 = 480$ (R: A)

$6^{\underline{4}} = 6 \cdot 5 \cdot 4 \cdot 3 = 30 \cdot 12 = 360$

3) B $\Rightarrow$ Ama



$8^{\underline{3}} = 8 \cdot 7 \cdot 6$

$= 14 \cdot 8 = 336$

$= 1680$

$7^{\underline{2}} = 7 \cdot 6 = 42 \cdot 2 = 84$

$336 - 84 = 252$ (R: D)

4) 3 R → _____
2 A
2 V
_____
RAAVV

$\binom{5}{2,2,1} = \frac{5!}{2!\cdot 2!} = \frac{5\cdot 4\cdot 3\cdot 2}{2\cdot 2} = 30$ R:A

5) (4,4) · (2,5) · (2,2) · (0,0)

$\binom{4}{2} + \binom{4}{2} \cdot \binom{4}{3}$

$= \frac{4\cdot 3}{2} \cdot \frac{4\cdot 3}{2} \cdot 4$

$12\cdot 12 = 144$ R:A

6) _ _ _ _

$\frac{4\cdot 3}{2} \cdot \frac{4\cdot 3}{2}$

$6\cdot 6\cdot 4 = 144$

$a+b+c+d=8$

$a=1 \quad b+c+d=8 \quad 8-5+1$

$\binom{3}{8} \quad \binom{10}{8} = \frac{10\cdot 9}{1\cdot 2}$

$= \frac{90}{2} = 45$

$a=2 \Rightarrow \quad b+c+d=7$

$\binom{3}{7} = \binom{9}{7} = \frac{9\cdot 8}{2} = 36$

$a=3$

$\binom{8}{6} = \frac{8\cdot 7}{2} = 28$

$81 + 28 = 109$ (A:E)